Linear Models Lecture 9: Probit and Poisson QMLE

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Binary Outcomes and the BLP

Recall from Lecture 2, the **best linear predictor** (BLP):

$$\beta = \arg\min_b \mathbb{E}\left[(Y - X'b)^2\right]$$

What happens when $Y \in \{0, 1\}$?
The conditional expectation function is now a probability:

$$m(x) = \mathbb{E}[Y \mid X = x] = P(Y = 1 \mid X = x)$$

The BLP still exists — OLS estimates β by solving the same normal equations:

$$\sum_{i=1}^{n} X_i(Y_i - X_i'\hat{\beta}) = 0$$

This is the **linear probability model** (LPM).

LPM: Consistent for Average Marginal Effects

Even if the true CEF $P(Y = 1 \mid X)$ is nonlinear, OLS estimates a useful object.
Recall the BLP–CEF distinction from Lecture 2:

- The BLP $X'\beta$ need not equal the CEF $m(x)$
- But β is consistent for the **average marginal effect** under mild conditions

Built-in heteroskedasticity (recall Lecture 6):
If $Y_i \in \{0, 1\}$, then

$$\mathrm{Var}(Y_i \mid X_i) = p(X_i)(1 - p(X_i))$$

where $p(X_i) = P(Y_i = 1 \mid X_i)$.
The variance depends on X_i by construction. The LPM **always** requires heteroskedasticity-robust standard errors (HC2).

Limitations of the Linear Probability Model

Problems:

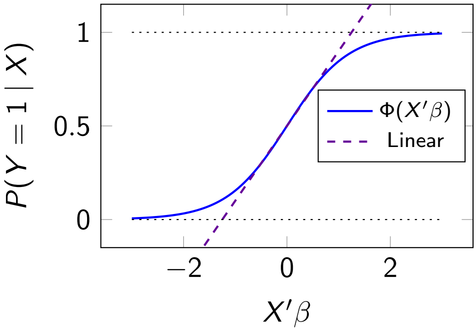
- Predictions outside $[0, 1]$
- Poor approximation in tails
- Marginal effects constant (may be unrealistic)

Motivation: A model that respects the $[0, 1]$ constraint.

Single-index model:

$$P(Y = 1 \mid X) = G(X'\beta)$$

where $G : \mathbb{R} \to [0, 1]$ is a known link function.



From LPM to Probit

Two common link functions:

Probit: $G = \Phi$ (standard normal CDF)
Logit: $G = \Lambda$ (logistic CDF: $e^x/(1+e^x)$)

Both map $\mathbb{R} \to (0,1)$, are symmetric about $1/2$, and produce nearly identical fitted values.

Why Probit?

- Natural latent variable interpretation ($\varepsilon \sim N(0,1)$)
- Connects to the normal distribution theory from Lectures 1 and 7
- Extends naturally to other nonlinear models (Poisson QMLE, later today)

Latent Variable Representation

Suppose there exists an unobserved variable:

$$Y_i^* = X_i'\beta + \varepsilon_i$$

Observed outcome:

$$Y_i = \mathbf{1}\{Y_i^* > 0\}$$

Assume $\varepsilon_i \sim N(0,1)$. Then:

$$\begin{aligned}
P(Y_i = 1 \mid X_i) &= P(Y_i^* > 0 \mid X_i) && \text{(definition of } Y_i\text{)} \\
&= P(X_i'\beta + \varepsilon_i > 0 \mid X_i) && \text{(substitute } Y_i^*\text{)} \\
&= P(\varepsilon_i > -X_i'\beta \mid X_i) && \text{(rearrange)} \\
&= 1 - \Phi(-X_i'\beta) && \text{(CDF of } \varepsilon_i\text{)} \\
&= \Phi(X_i'\beta) && \text{(symmetry of } N(0,1)\text{)}
\end{aligned}$$

This is the **Probit model**.

Where does Y_i^* come from? Random Utility

Think of Y_i^* as the difference in utility between two choices.
Agent i chooses between alternatives 0 and 1, with utilities:

$$U_{i1} = X_i'\beta_1 + \varepsilon_{i1}, \qquad U_{i0} = X_i'\beta_0 + \varepsilon_{i0}$$

Agent picks alternative 1 whenever $U_{i1} > U_{i0}$:

$$Y_i = 1 \iff X_i'(\beta_1 - \beta_0) + (\varepsilon_{i1} - \varepsilon_{i0}) > 0$$

Define $\beta \equiv \beta_1 - \beta_0$ and $\varepsilon_i \equiv \varepsilon_{i1} - \varepsilon_{i0}$:

$$Y_i^* = X_i'\beta + \varepsilon_i, \qquad Y_i = \mathbf{1}\{Y_i^* > 0\}$$

- Only the *difference* $\beta_1 - \beta_0$ is identified — the levels are not.
- ε_{ij} iid $N(0, \frac{1}{2}) \Rightarrow \varepsilon_i \sim N(0,1)$: **probit**.
- ε_{ij} iid Type-I extreme value $\Rightarrow \varepsilon_i$ logistic: **logit**.

Identification and Scale Normalization

Now let $\varepsilon_i \sim N(0, \sigma^2)$ with σ unknown. Since $\varepsilon_i/\sigma \sim N(0, 1)$:

$$P(Y_i = 1 \mid X_i) = P\big(\varepsilon_i > -X_i'\beta \mid X_i\big) = P\left(\frac{\varepsilon_i}{\sigma} > -\frac{X_i'\beta}{\sigma}\right) = \Phi\left(\frac{X_i'\beta}{\sigma}\right)$$

The likelihood depends on (β, σ) only through the ratio β/σ: for any $c > 0$, (β, σ) and $(c\beta, c\sigma)$ yield identical $P(Y_i = 1 \mid X_i)$ for every X_i.

Normalization: Set $\text{Var}(\varepsilon_i) = 1$. The "$\beta$" we estimate is really β/σ — identified only up to scale.

Scale Is to Probit What the Intercept Is to OLS

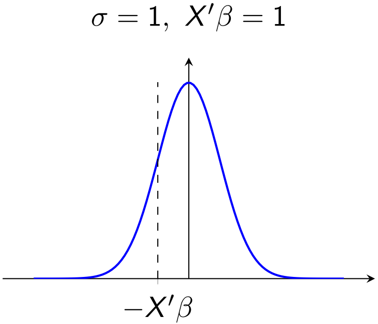
The non-identification is the scale analog of a familiar location problem.

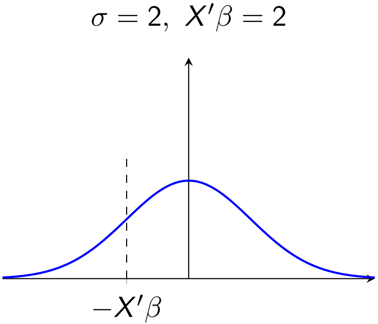
	OLS (location)	**Probit (scale)**
Model	$Y = \alpha + X'\beta + \varepsilon$	$Y^* = X'\beta + \varepsilon,\ Y = \mathbf{1}\{Y^* > 0\}$
Data identify only	$\alpha + E[\varepsilon]$	β/σ
Absorbed by	intercept α	coefficients β
Normalization	$E[\varepsilon] = 0$	$\text{Var}(\varepsilon) = 1$

Shifting the ε distribution left/right moves the intercept by the same amount; stretching it wider/narrower scales β by the same factor. In each case a single nuisance parameter rides along with a coefficient, and we normalize it away.

Picture: Same $P(Y = 1 \mid X)$, Different (β, σ)

Two probit DGPs. The data see only the shaded tail $P(\varepsilon_i > -X_i'\beta)$.





Both shaded areas equal $\Phi(X_i'\beta/\sigma) = \Phi(1)$. Doubling σ and β together leaves the choice probability unchanged for every X_i — so the likelihood cannot distinguish them.

Marginal Effects

In the linear model: $\partial\mathbb{E}[Y \mid X]/\partial x_j = \beta_j$.
In the Probit model:

$$\frac{\partial P(Y = 1 \mid X)}{\partial x_j} = \phi(X'\beta) \cdot \beta_j$$

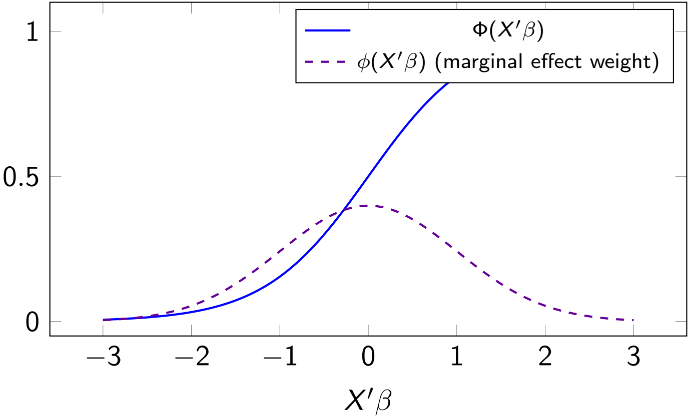
The marginal effect **depends on** X through $\phi(X'\beta)$.

Average Marginal Effect (AME):

$$\widehat{\text{AME}}_j = \frac{1}{n}\sum_{i=1}^{n} \phi(X_i'\hat{\beta})\,\hat{\beta}_j$$

- The AME averages over the observed distribution of X
- This is the natural analog of the OLS coefficient for binary outcomes
- Computing the AME requires the **delta method** (Lecture 10) for standard errors

Marginal Effects: Graphical Intuition



Marginal effects are largest near $X'\beta = 0$ (where the CDF is steepest) and vanish in the tails.

From Bernoulli to the Probit Likelihood

Each binary outcome is a **Bernoulli trial**. For a single $Y \in \{0, 1\}$ with success probability p:

$$f(y \mid p) = p^y (1-p)^{1-y}$$

No covariates: if $Y_1, \ldots, Y_n$ are iid Bernoulli(p), then $S = \sum_i Y_i \sim \text{Binomial}(n, p)$, and

$$L(p) = \binom{n}{S} p^S (1-p)^{n-S}$$

With covariates: each trial has its own success probability,

$$p_i = P(Y_i = 1 \mid X_i) = \Phi(X_i' \beta)$$

Under conditional independence, the joint density is the product of n heterogeneous Bernoullis — no binomial coefficient because the trials are distinguishable through X_i.

Likelihood for the Probit Model

Viewing that joint density as a function of β gives the **likelihood**:

$$L_n(\beta) = \prod_{i=1}^{n} \Phi(X_i'\beta)^{Y_i} \left(1 - \Phi(X_i'\beta)\right)^{1-Y_i}$$

Log-likelihood:

$$\ell_n(\beta) = \sum_{i=1}^{n} \left[Y_i \log \Phi(X_i'\beta) + (1 - Y_i) \log\left(1 - \Phi(X_i'\beta)\right)\right]$$

We estimate $\hat{\beta}_{\text{MLE}} = \arg\max_\beta \ \ell_n(\beta)$.

Score Function

Let $\phi(\cdot)$ denote the standard normal pdf, $\Phi(\cdot)$ the CDF.
The individual score contribution:

$$s_i(\beta) = \frac{\partial \ell_i}{\partial \beta} = \frac{\phi(X_i'\beta)}{\Phi(X_i'\beta)(1-\Phi(X_i'\beta))} \left(Y_i - \Phi(X_i'\beta)\right) X_i$$

The total score:

$$S_n(\beta) = \sum_{i=1}^{n} s_i(\beta) = \sum_{i=1}^{n} w_i \left(Y_i - \Phi(X_i'\beta)\right) X_i$$

where $w_i = \phi(X_i'\beta)/[\Phi(X_i'\beta)(1-\Phi(X_i'\beta))]$.

Setting $S_n(\hat{\beta}) = 0$ defines the MLE. This is a **nonlinear** system — no closed-form solution.

Score in Practice: What `glm()` Solves

```
probit <- glm(Y ~ X1 + X2, data = dta,
              family = binomial(link = "probit"))
# Compute the score at the MLE (should be ~0)
xb  <- predict(probit, type = "link")
w   <- dnorm(xb) / (pnorm(xb) * (1 - pnorm(xb)))
res <- dta$Y - pnorm(xb)           # Y_i - Phi(X'beta)
score <- colSums(w * res * model.matrix(probit))
round(score, 10)
#  (Intercept)        X1        X2
#            0         0         0
```

The score is what `glm()` drives to zero. It is the nonlinear analog of the OLS normal equations $\boldsymbol{X}'\hat{\boldsymbol{e}} = 0$.

Score as Moment Condition

Key insight: Under correct specification, $\mathbb{E}[s_i(\beta_0)] = 0$.

Compare three estimating equations side by side:

Model	Estimating Equation	Mean Function
OLS	$\sum_i X_i(Y_i - X_i'\beta) = 0$	$X'\beta$
Probit MLE	$\sum_i w_i(Y_i - \Phi(X_i'\beta))\, X_i = 0$	$\Phi(X'\beta)$
Poisson QMLE	$\sum_i X_i(Y_i - \exp(X_i'\beta)) = 0$	$\exp(X'\beta)$

All three are **moment conditions**: find θ s.t. $\frac{1}{n}\sum_i g(W_i, \theta) = 0$ (GMM).

Iterative Estimation: Newton–Raphson

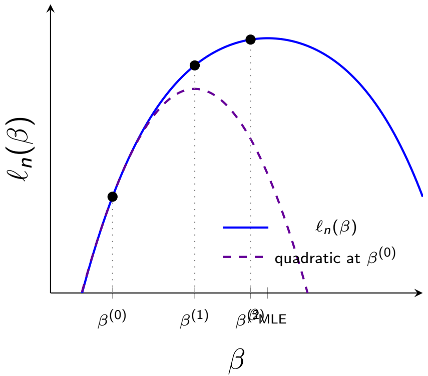
Probit MLE has no closed form — iterate.
Quadratic Taylor expansion of ℓ_n at $\beta^{(t)}$:

$$\ell_n(\beta) \approx \ell_n^{(t)} + S_n^{(t)}\Delta + \frac{1}{2}H_n^{(t)}\Delta^2$$

with $\Delta = \beta - \beta^{(t)}$.
Jump to its max:

$$\boxed{\beta^{(t+1)} = \beta^{(t)} - \left[H_n^{(t)}\right]^{-1} S_n^{(t)}}$$

Repeat until convergence. Fast (3–6 steps) because ℓ_n is concave.



Each step peaks the local quadratic and jumps there. `glm()` does this via **Fisher scoring**.

Newton–Raphson in Practice: Watching `glm()` Iterate

```
probit <- glm(Y ~ X1 + X2, data = dta,
              family = binomial(link = "probit"),
              control = list(trace = TRUE))
# Deviance = -2 * log-likelihood (lower = better)
#   Deviance = 134.60 Iterations - 1
#   Deviance = 130.28 Iterations - 2
#   Deviance = 130.21 Iterations - 3
#   Deviance = 130.21 Iterations - 4   <- converged
```

- Each iteration applies the Newton–Raphson update $\beta^{(t+1)} = \beta^{(t)} - H^{-1}S$.
- Convergence is typically fast (3–6 iterations) because the log-likelihood is concave.
- If `glm()` warns "algorithm did not converge," the score could not be driven to zero — a sign of separation or misspecification.

Why Not Just Use OLS?

	LPM (OLS)	Probit (MLE)
Closed-form solution	Yes	No
Predictions in $[0, 1]$	Not guaranteed	Yes
Consistent for β	—	Yes (if correctly specified)
Consistent for AME	Yes (always)	Yes (if correctly specified)
Robust to misspecification	Yes (BLP always exists)	Requires sandwich SEs
Efficiency	Lower	Higher (if correct)

Practical guidance: LPM is a robust baseline. Probit gains efficiency if the model is correct, but misspecification has consequences.

From Binary to Nonnegative Outcomes

Probit handles $Y \in \{0, 1\}$ by passing $X'\beta$ through Φ.
What about $Y \geq 0$? Counts, dollar amounts, rates, corner solutions.

Same problem as LPM: a linear model $\mathbb{E}[Y \mid X] = X'\beta$ can predict **negative values** for an outcome that cannot be negative.

Solution: the exponential conditional mean.

$$\mathbb{E}[Y \mid X] = \exp(X'\beta)$$

- Always positive
- Natural for counts, corner solutions, and any $Y \geq 0$
- The analog of $\Phi(X'\beta)$ for binary outcomes

The Poisson Distribution

A random variable Y is **Poisson** with rate $\lambda > 0$ if it takes nonnegative integer values with probability mass

$$P(Y = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \qquad k = 0, 1, 2, \dots$$

Properties:

- Support: $\{0, 1, 2, \dots\}$ — a counting distribution
- Mean: $\mathbb{E}[Y] = \lambda$
- Variance: $\text{Var}(Y) = \lambda$ (**equidispersion**: variance equals mean)

The single parameter λ pins down every moment — a very restrictive assumption.

Why "Poisson" Regression?

Poisson regression is conventionally written as a **log-linear** model for the conditional mean:

$$\log \mathbb{E}[Y_i \mid X_i] = X_i'\beta \quad \Longleftrightarrow \quad \mathbb{E}[Y_i \mid X_i] = \exp(X_i'\beta)$$

The full model assumes the conditional distribution is Poisson with that rate:

$$\lambda_i = \exp(X_i'\beta), \qquad Y_i \mid X_i \sim \text{Poisson}(\lambda_i)$$

which gives $\mathbb{E}[Y_i \mid X_i] = \lambda_i$ and automatically respects $\lambda_i > 0$.

We will see the name outlives the distributional assumption: consistency for β needs only the log-linear mean $\log \mathbb{E}[Y \mid X] = X'\beta$. Y need not be integer, and the variance need not equal the mean.

Why $\exp(X'\beta)$? Invariance

Positivity is not the only reason. The exponential mean is natural when the **mechanism** implies scale invariance:

- **Proportional change** (population growth, compounding returns): effects should be stable under rescaling $\rightarrow$ a semi-elasticity, not an absolute change
- **Event arrival** (conflicts, patent filings): expected counts should add when you double the observation window
- Both imply $\mathbb{E}[Y \mid X] = \exp(X'\beta)$, not $X'\beta$

The choice between $X'\beta$ and $\exp(X'\beta)$ is not just about the data format—it reflects a claim about what transformation of the world leaves the effect unchanged.

The Poisson Quasi-Log-Likelihood

If $Y_i \mid X_i \sim \text{Poisson}(\lambda_i)$ with $\lambda_i = \exp(X_i'\beta)$, the log-likelihood for one observation is

$$\ell_i(\beta) = Y_i \cdot X_i'\beta - \exp(X_i'\beta) - \log(Y_i!)$$

Why "quasi"? We maximize this ℓ_i even when we do *not* believe Y is Poisson. It is a true likelihood only under the full distributional assumption; otherwise it is simply an **objective function** borrowed from the Poisson form.

Score:

$$s_i(\beta) = \big(Y_i - \exp(X_i'\beta)\big)X_i \quad \Longrightarrow \quad \boxed{\sum_{i=1}^{n} X_i\big(Y_i - \exp(X_i'\beta)\big) = 0}$$

Compare to OLS: $\sum X_i(Y_i - X_i'\beta) = 0$. Same structure — a **nonlinear moment condition**, with no observation-specific weights (unlike probit).

QMLE: Consistency Without the Distribution

Key Result (Gourieroux et al. 1984; Wooldridge 2010, 2023)

Poisson QMLE is **consistent for** β whenever $\mathbb{E}[Y \mid X] = \exp(X'\beta)$ is correctly specified. Y need **not** be Poisson. It need not even be integer-valued.

This is **quasi-MLE**: use a likelihood as an estimation device, not as a distributional assumption.

What can Y be?

- Integer counts (patents, murders, votes)
- Continuous nonneg (income, expenditure, trade flows)
- Corner solutions (zero with positive mass, then continuous)
- Fractions in $[0, \infty)$

All that matters is that the conditional mean is exponential.

Interpreting Poisson QMLE Coefficients

Because $\mathbb{E}[Y \mid X] = \exp(X'\beta)$:

$$\frac{\mathbb{E}[Y \mid x_j + 1, \mathbf{x}_{-j}]}{\mathbb{E}[Y \mid x_j, \mathbf{x}_{-j}]} = \exp(\beta_j)$$

β_j is a **semi-elasticity**: a one-unit increase in x_j changes $\mathbb{E}[Y \mid X]$ by approximately $100 \times \beta_j$ percent.

Model	β_j **measures**	**Interpretation**
OLS	$\Delta\mathbb{E}[Y \mid X]$	Absolute change
Probit	$\phi(X'\beta) \cdot \beta_j$	Needs ME transformation
Poisson QMLE	$\approx 100\beta_j\%$ change	Semi-elasticity

Poisson coefficients are directly interpretable — no marginal effect computation needed.

Sandwich SEs Are Mandatory

The Poisson variance assumption: $\text{Var}(Y \mid X) = \mathbb{E}[Y \mid X]$.

This is almost **never correct** in practice. With overdispersion:

$$\text{Var}(Y \mid X) \gg \mathbb{E}[Y \mid X]$$

When $\text{Var}(Y \mid X) \neq \mathbb{E}[Y \mid X]$, model-based SEs are wrong.

Practical Rule

Always use sandwich standard errors with Poisson QMLE. The model-based SEs from `summary(glm(...))` are not valid. This is the same sandwich from Lectures 5–6, applied to a nonlinear model.

Contrast with probit: sandwich SEs are *optional* under correct specification, **mandatory** for Poisson QMLE.

The LEF Framework (Wooldridge 2023)

Linear exponential family (LEF) distributions provide a unified framework:

Cond. Mean	LEF Density	Outcome Type	R Code
Linear	Normal	Any Y	`lm()`
Logistic	Bernoulli	Binary / fractional	`glm(, binomial)`
Exponential	Poisson	$Y \geq 0$, no upper bound	`glm(, poisson)`

In each case, consistency requires only that the **conditional mean** is correctly specified—not the full distribution. Use sandwich SEs in all cases.

Applied Workflow

When estimating models for nonnegative outcomes:

1. Estimate OLS (linear) as a baseline
2. Estimate Poisson QMLE (exponential mean) for $Y \geq 0$
3. Compare estimates as a **robustness check**
4. Use sandwich SEs in all cases
5. If estimates diverge, investigate functional form

This extends naturally to panel data (DiD with nonneg outcomes).

Probit in R: glm()

```
# Linear Probability Model
lpm <- lm(Y ~ X1 + X2, data = dta)

# Probit Model
probit <- glm(Y ~ X1 + X2, data = dta,
              family = binomial(link = "probit"))

# Compare coefficients
cbind(LPM = coef(lpm), Probit = coef(probit))
```

- `glm()` uses Fisher scoring (iteratively reweighted least squares)
- `family = binomial(link = "probit")` specifies $G = \Phi$
- For logit: `family = binomial(link = "logit")`

Marginal Effects in R

Probit coefficients $\hat{\beta}_j$ are **not** marginal effects. Compute them manually:

```
# Average Marginal Effect (AME)
xb <- predict(probit, type = "link")  # X'beta-hat
ame <- mean(dnorm(xb)) * coef(probit)
ame
```

Or using the `margins` package:

```
library(margins)
summary(margins(probit))
```

The AME is directly comparable to the LPM coefficient — both estimate $\mathbb{E}[\partial P/\partial x_j]$.

Robust Standard Errors for Probit

Same `sandwich` tools from Lecture 6:

```
library(sandwich)
library(lmtest)

# Default (model-based) SEs
coeftest(probit)

# Robust (sandwich) SEs
coeftest(probit, vcov = vcovHC(probit, type = "HC1"))
```

- Model-based SEs rely on correct specification of Φ as the link
- Sandwich SEs are valid even under misspecification (QMLE)
- If model-based and robust SEs differ substantially, this signals possible misspecification

Poisson QMLE in R: `glm()`

```
# Poisson QMLE for nonnegative Y
pois <- glm(Y ~ X1 + X2, data = dta,
            family = poisson)

# MANDATORY: sandwich SEs (model-based are wrong)
coeftest(pois, vcov = vcovHC(pois, type = "HC1"))
```

```
# Coefficient interpretation
exp(coef(pois))  # multiplicative effects
```

- Works even if `Y` is continuous (non-integer) — R warns, but estimates are valid
- `exp(coef(pois)["X1"])` = multiplicative effect of one-unit increase in `X1`
- Compare to OLS: `lm(Y ~ X1 + X2)` as a robustness check

Comparison Table

	OLS	LPM	Probit	Poisson QMLE
Outcome	Any	Binary	Binary	$Y \geq 0$
CEF	$X'\beta$	$X'\beta$	$\Phi(X'\beta)$	$\exp(X'\beta)$
β_j means	ΔY	ΔP	needs ME	$\approx \%\Delta Y$
Moment cond.	$\sum Xe = 0$	$\sum Xe = 0$	$\sum ws = 0$	$\sum X(Y - e^{X'\beta}) = 0$
Sandwich SEs	Rec.	Mandatory	Optional	Mandatory
Invariance	Translation	Translation	Monotone latent	Scale

The GMM generalization: All of these are special cases of

$$\mathbb{E}[g(W_i, \theta_0)] = 0$$

Find $\hat{\theta}$ such that $\frac{1}{n}\sum_{i=1}^{n} g(W_i, \hat{\theta}) \approx 0$. This is the **Generalized Method of Moments**.

Key Takeaways and Looking Ahead

Today:

- Binary outcomes $\Rightarrow$ Probit; nonneg outcomes $\Rightarrow$ Poisson QMLE
- LPM and OLS remain useful baselines for comparison
- Probit: β_j needs marginal effect transformation; Poisson: $\beta_j \approx$ semi-elasticity
- All score equations are **moment conditions** — nonlinear GMM
- Poisson QMLE: consistent if $\mathbb{E}[Y \mid X] = \exp(X'\beta)$, regardless of the distribution of Y
- Sandwich SEs: optional for probit (under correct spec), mandatory for Poisson QMLE
- Always compare linear and nonlinear estimates as a robustness check

Next (Lecture 10): The formal asymptotic tools — WLLN, CLT, delta method — that justify everything we did today.